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IE 2004 Computer Networks

02 – Protocol architecture

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Lesson outline

- What are Protocols?
- Need for protocols in networking
- Protocol layering and Encapsulation
- TCP/IP protocol stack
- Comparison of OSI and TCP/IP

What's a protocol?

Human protocols:

- "What is the time?"
- "Any questions?"
- Introducing a friend to another

Network protocols:

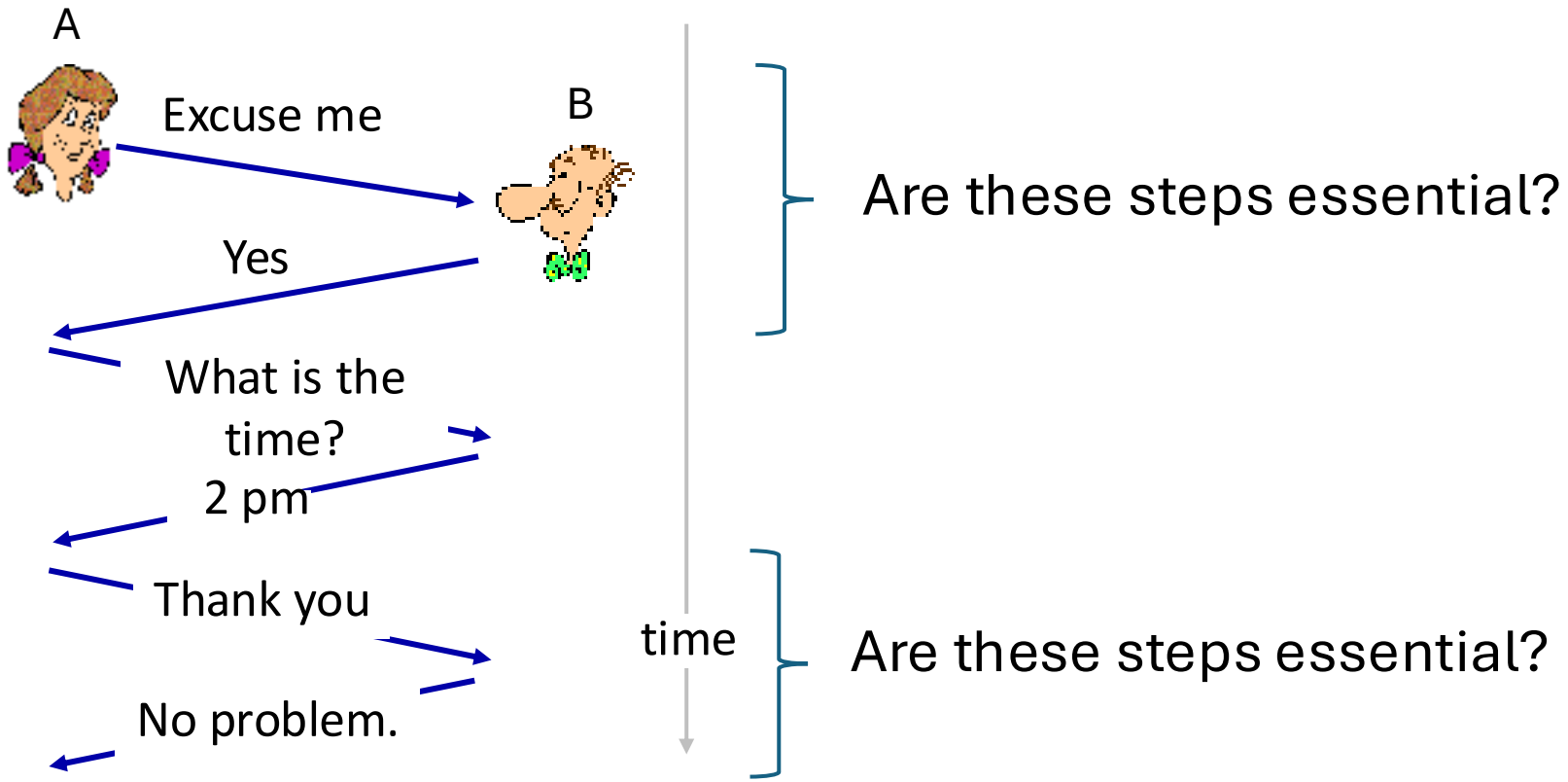
- computers (devices) rather than humans
- all communication activity in Internet governed by protocols

Protocols define

- format of messages
- order of messages sent and received among network entities
- actions taken after a message is sent or received

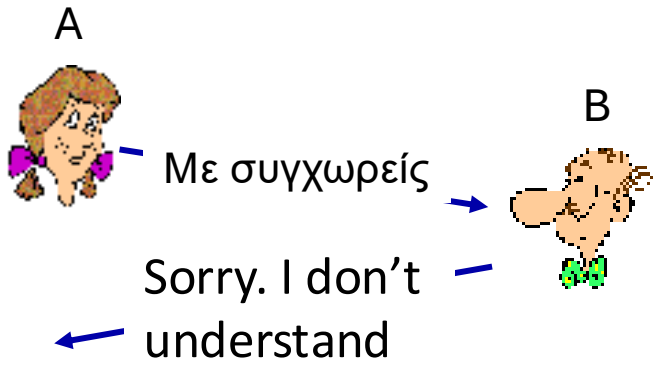
Example: What is the time?

Example human protocol



Example: What is the time?

Example human protocol



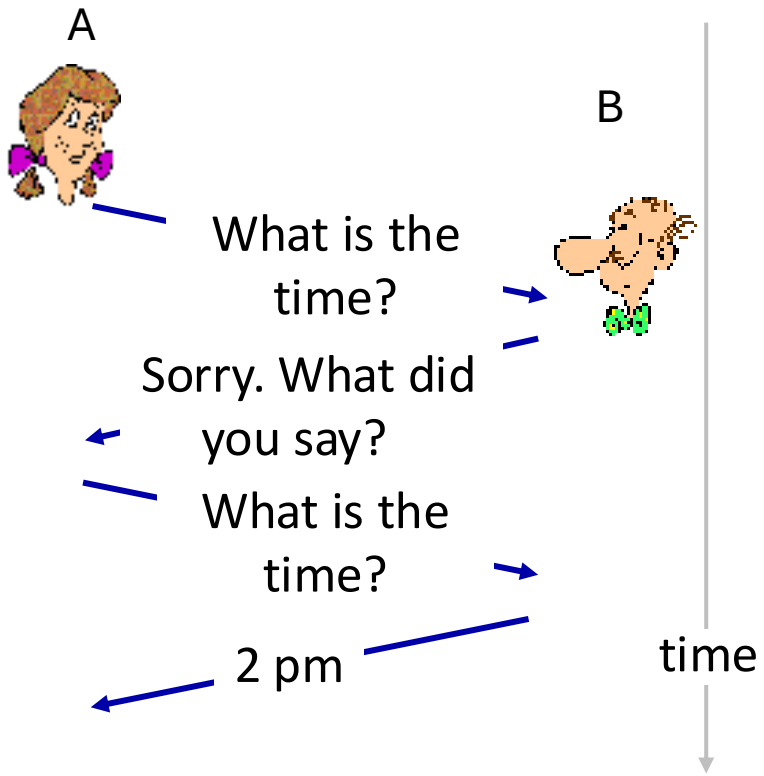
- In this case the communication task is incomplete
  - cannot proceed to next steps

Are these steps essential?

- Not essential if both are speaking the same language.
- But safer to have them
  - to make sure that both can understand each other
  - to make sure that A has got B's attention

Example: What is the time?

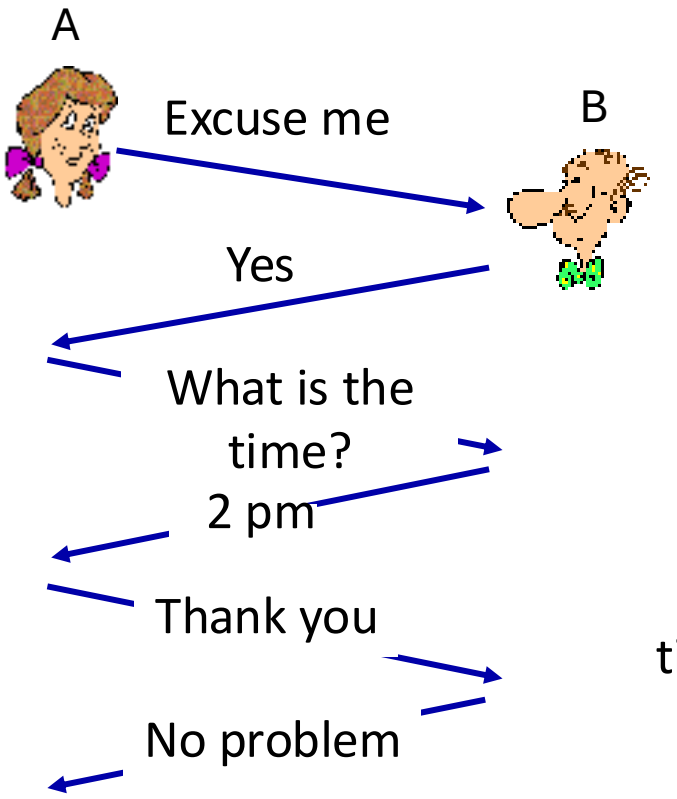
Example human protocol



A did not have B's attention to ask the question.

Example: What is the time?

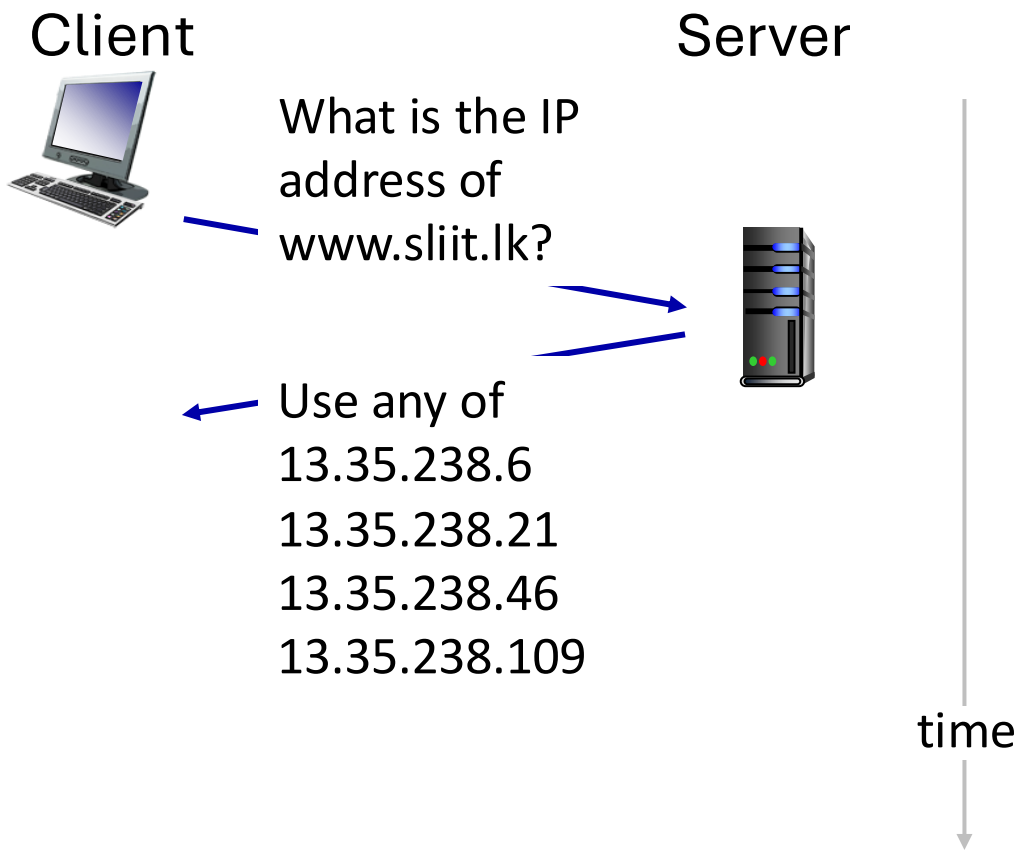
Example human protocol



These two steps are optional
- No “Thank you” and “No problem”
- Just “Thank you”
- “Thank you” and then “No problem”

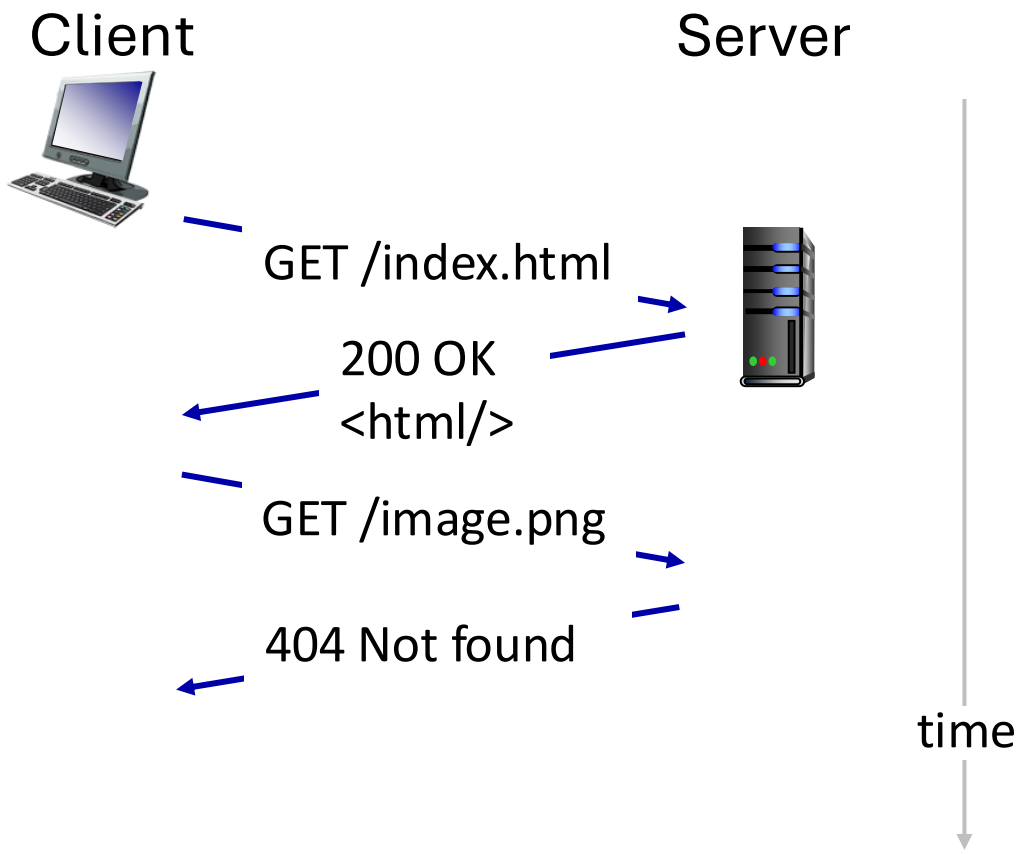
Example: DNS request

- A client asking a Domain Name System (DNS) server about the numerical address of a domain name

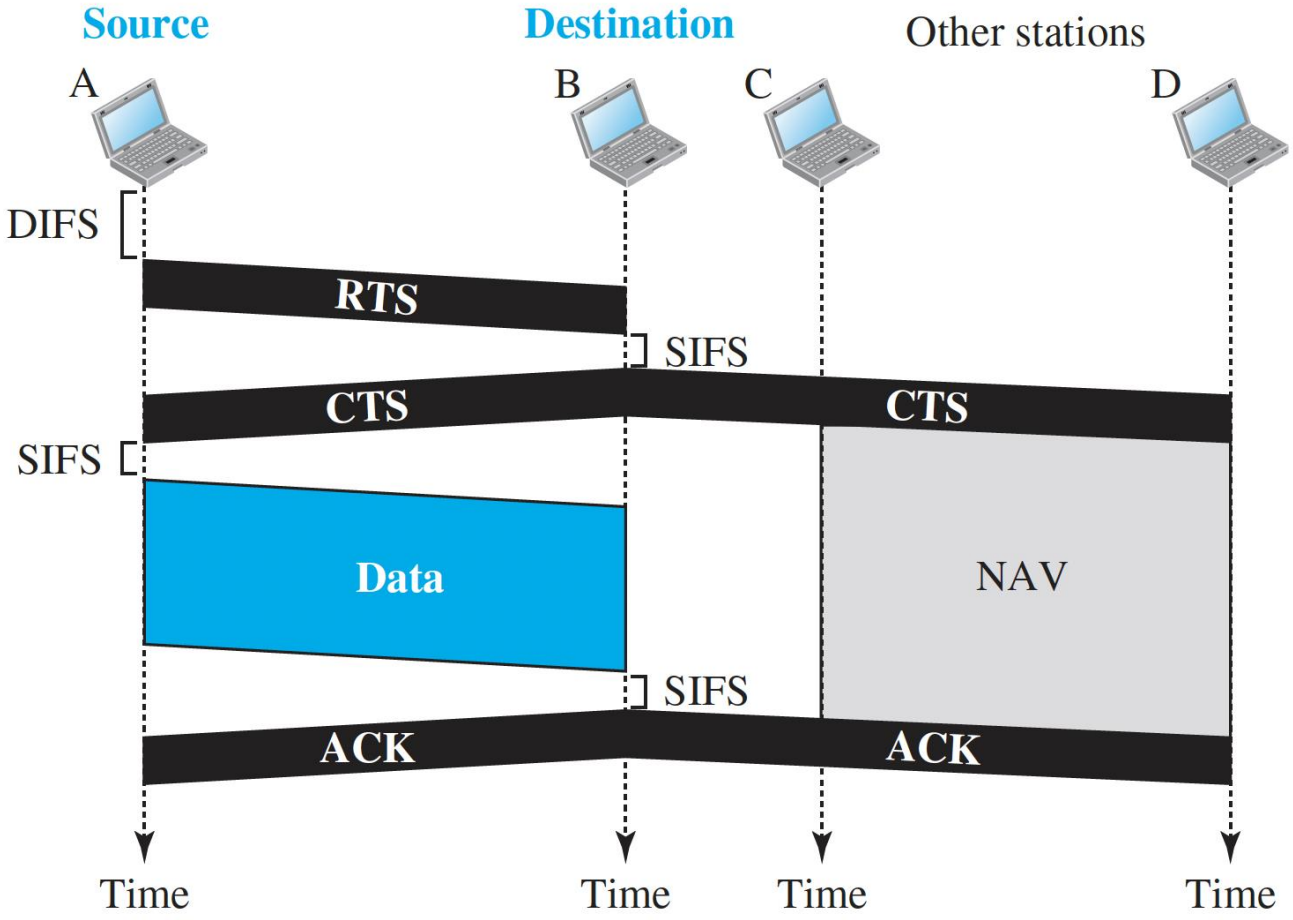


Example: Web page request

- A web browser(client) accessing a web page on a web server using HTTP (Hyper Text Transfer Protocol)



Example: Four frame exchange in Wi-Fi



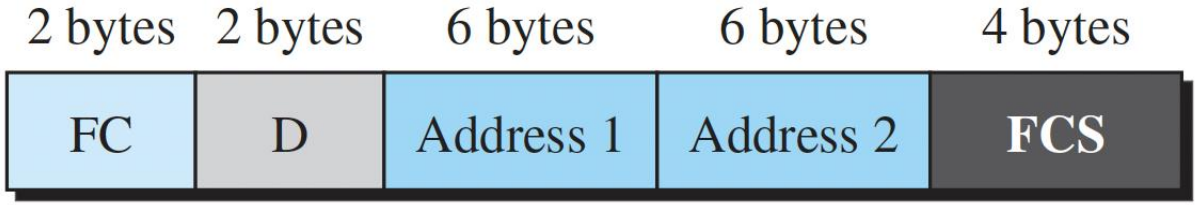
1. Source issues a Request to Send (RTS) frame
2. Destination responds with Clear to Send (CTS)
3. After receiving CTS, source transmits data
4. Destination responds with ACK

Example: Four frame exchange in Wi-Fi (ctd)

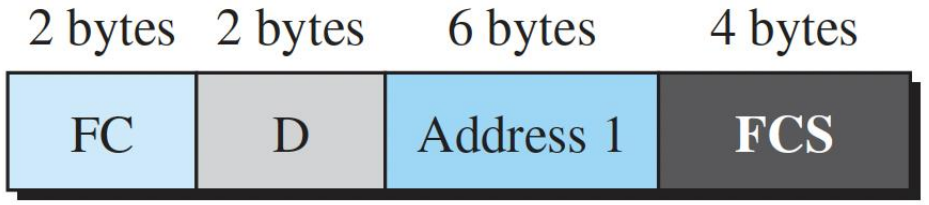
- Sender (A) waits until no one else is “speaking”
- Continues to wait for a specified time period (DIFS = 34 µs)
- If no one is “speaking” even after that time period, it sends a Request to Send (RTS) to the intended recipient
- After receiving the RTS, the recipient (B) waits for a time period (SIFS = 16 µs)
- Recipient sends Clear to Send (CTS) to the sender
- After receiving CTS, the sender waits for a time period (SIFS) and sends the data
- Other stations “keep quiet” for a time period specified in RTS/CTS messages
- After receiving the data, the recipient waits for a time period (SIFS) and sends an Acknowledgment
- If no Acknowledgement is received within a timeout, sender starts all over again

Example: Four frame exchange in Wi-Fi (ctd)

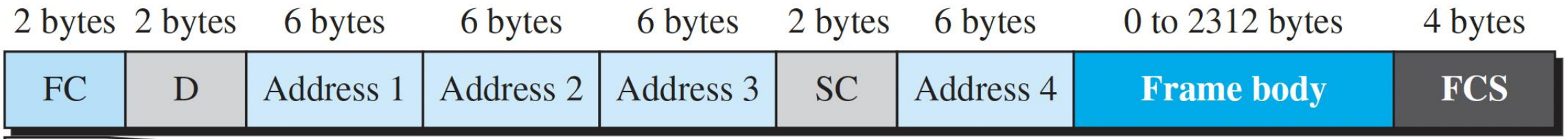
- Message format for Wi-Fi

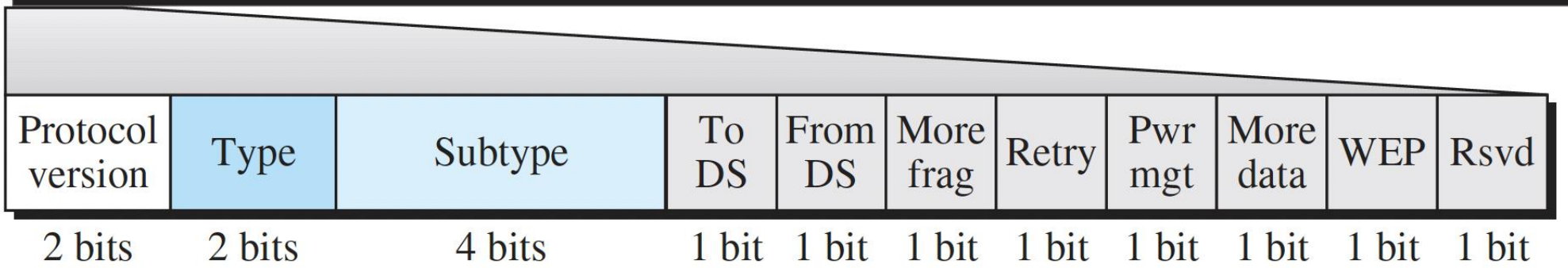


RTS



CTS or ACK





DATA

Revisit: What is a protocol?

- Protocols define
  - format of messages
  - order of messages sent and received among network entities
  - actions taken when a message is sent or received
  - actions taken if an expected message is not received
- Some messages are essential, while others may be optional or required in certain situations
  - In “What is the time?”, it is ok if A does not say “Thank you”
- Violating the protocol can make the communication difficult or impossible
- The messages exchanged are called protocol data units (PDUs)

Exercises

- What are possible variations in “What is the time?” protocol?
  - What happens if the expected response is not received?
- Identify the messages and actions involved in another human to human communication, like
  - Teacher asking “Do you have any questions?” in a lecture
  - Buying a lunch packet from the canteen
  - Consulting a doctor
- In the communication above, identify actions that can be taken if the expected response / action is not received within a certain time

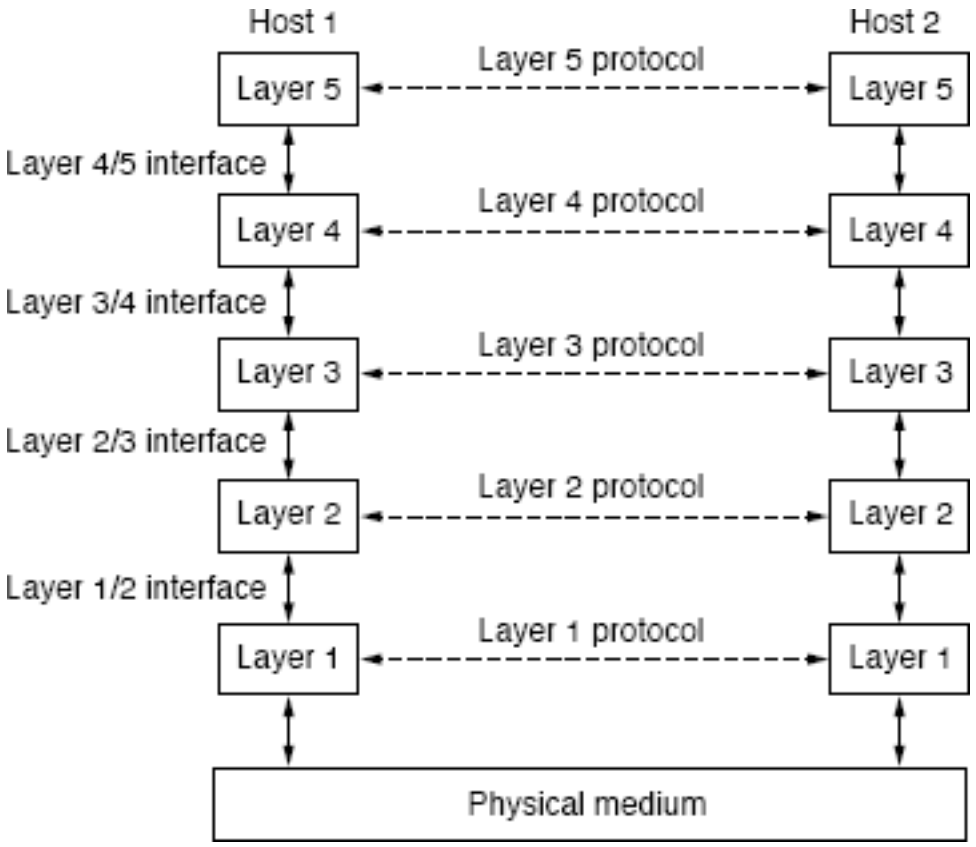
Review questions

- What does a protocol include?
- What is a protocol data unit (PDU)?
- In protocols, if an expected response is not received within a pre-defined time after sending the message, a timeout occurs.
  - Give two possible scenarios which can result in the expected response not arriving at the sender.
  - What are possible actions that the sender can take when a timeout occurs?

Need for a protocol architecture

- Communicating with another device is a complex task
- Better if this big task can be broken into a number of smaller subtasks
- Each subtask can be performed by a layer (hardware and/or software) specialized to do that subtask
- Each layer (except layer 5) provides services to the layer above
  - Each layer (except layer 1) uses the services provided by the layer below

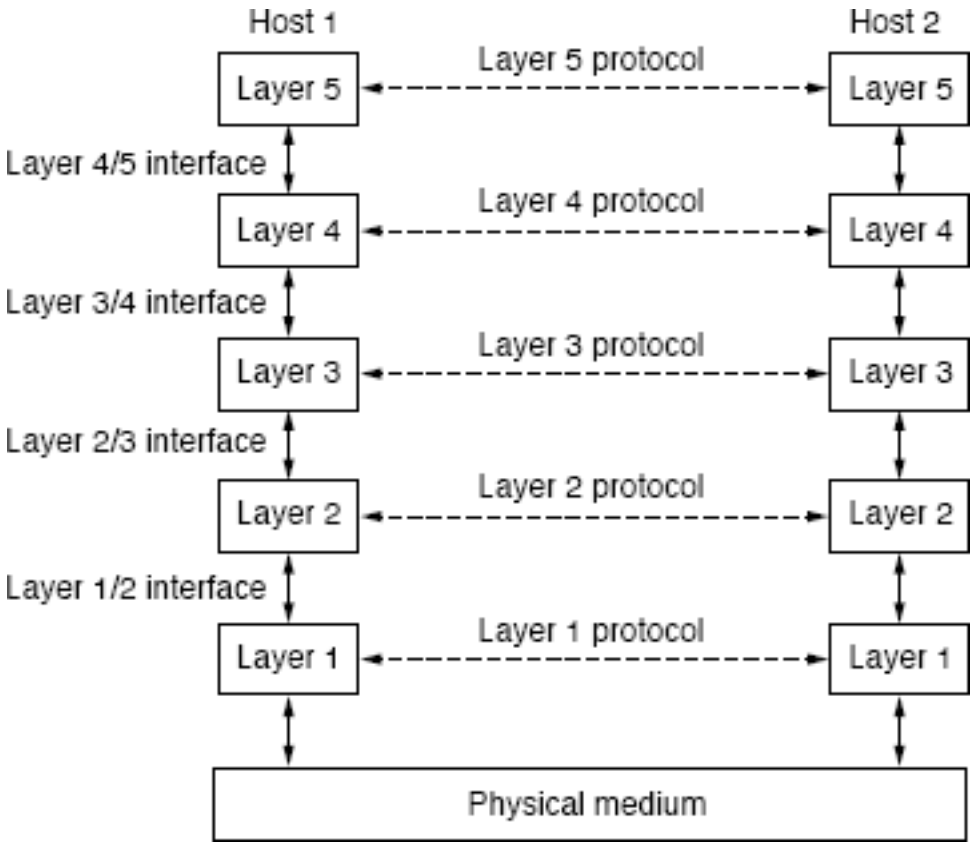
An example 5-layer architecture



Need for a protocol architecture (2)

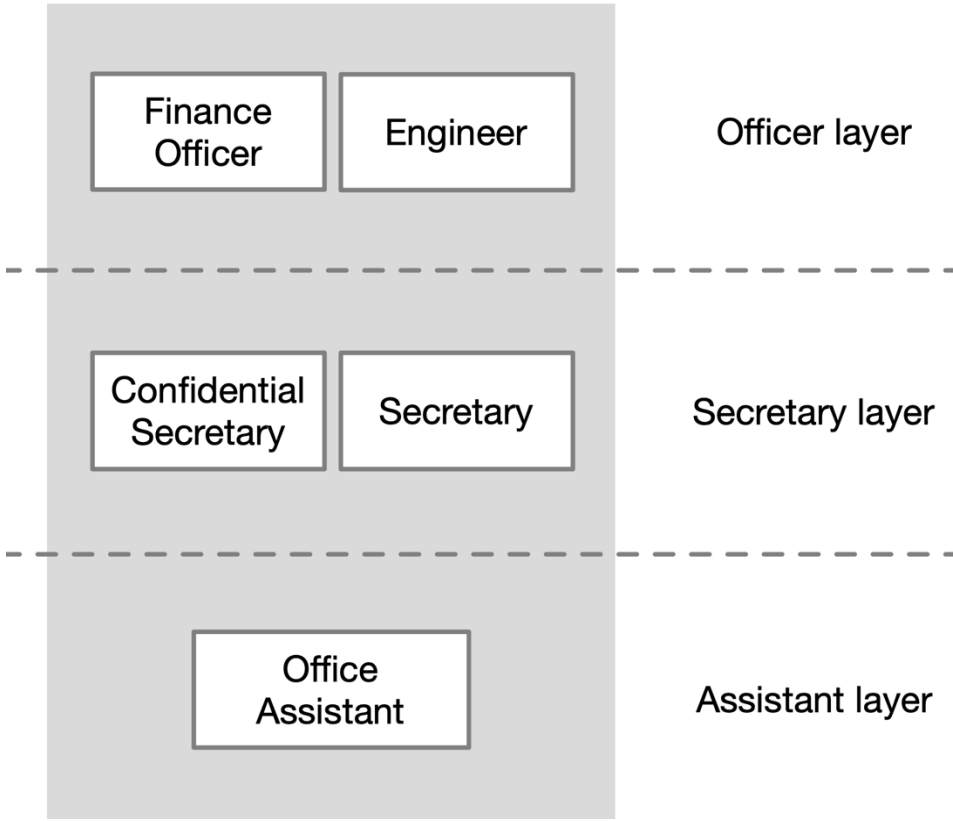
- Each layer can be modified independently, as long as services provided to upper layer and services expected from lower layer do not change
  - i.e. the interfaces between the layers do not change
- Peer layers at the two hosts communicate using a protocol specific to that layer
  - Can be changed without affecting the upper or lower layer protocols

An example 5-layer architecture



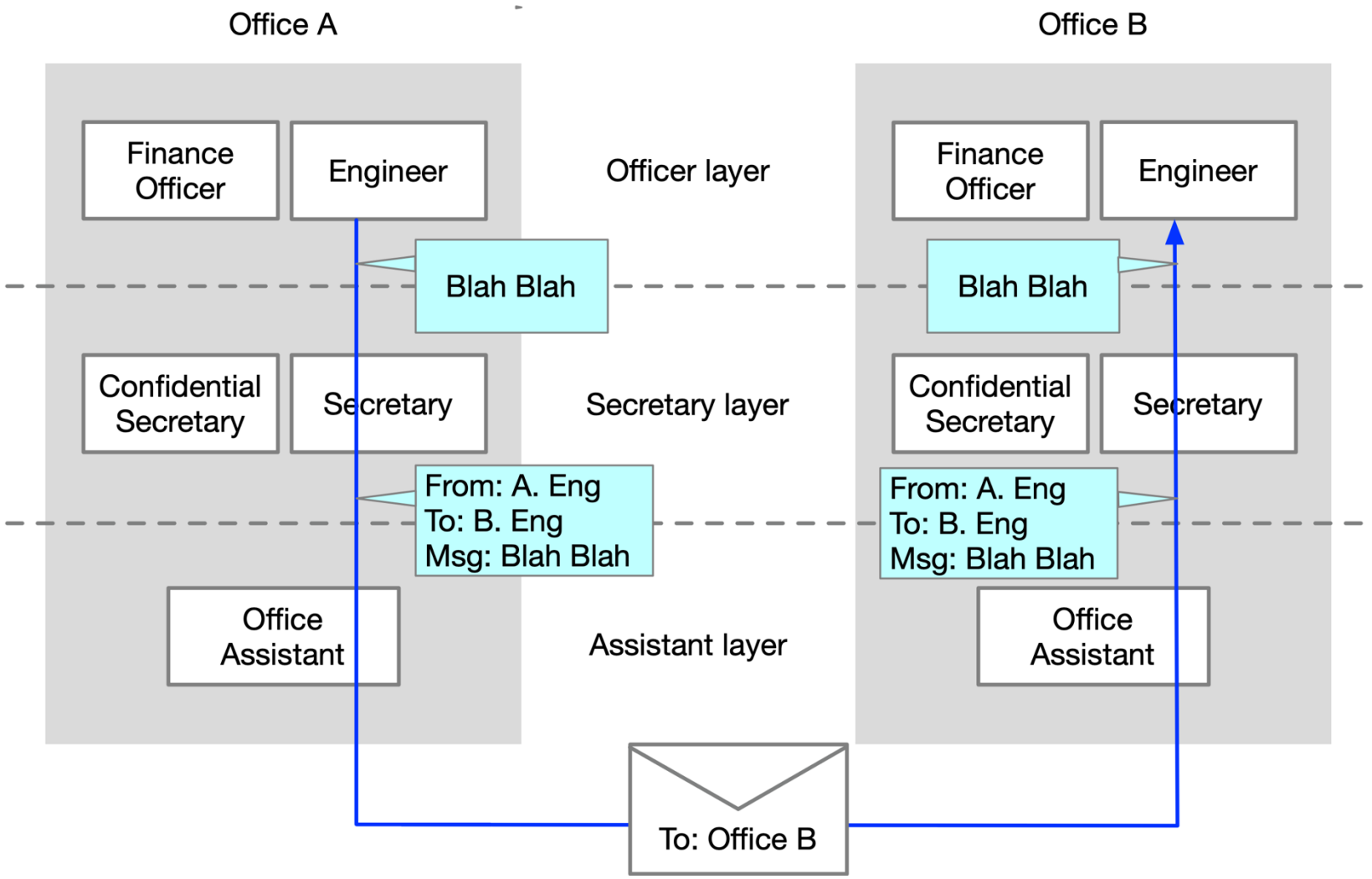
Protocol layering: an analogy

- Consider an office consisting of the following personnel
  - Officer grade: Engineer and Finance officer
  - Secretary grade: A confidential secretary to handle confidential matters and a secretary to handle other matters
  - Office assistant: Sends documents given by a secretary to the destination, and passes received documents to the appropriate secretary
- Each layer provides services to the upper layer and uses the services provided by the lower layer
- It is obvious that dividing the tasks in the office into subtasks and using specialized persons to do each subtask is better than having a single person doing all the tasks
  - Each person can be changed independently without affecting others in the office



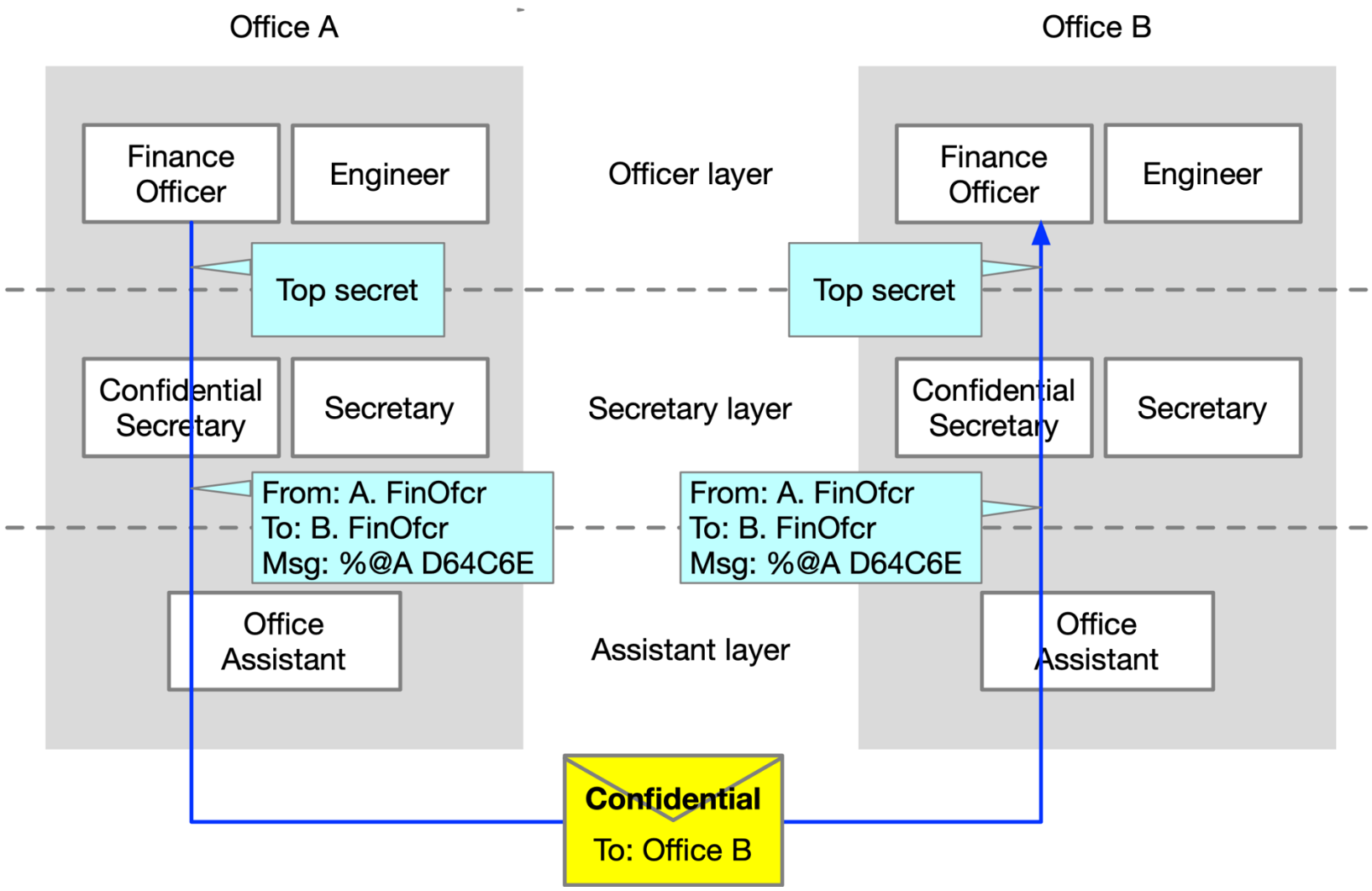
Protocol layering analogy – scenario 1

- Engineer in Office A communicating with engineer in office B
- Using the services of the normal secretary
- Note that things are added to/removed from the message at each layer



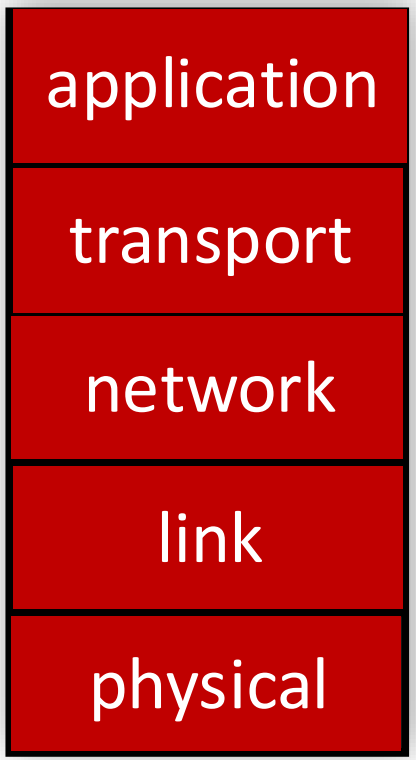
Protocol layering analogy – scenario 2

- Finance officer in Office A communicating with finance officer in office B
- Message requires confidentiality
  - Use the services of confidential secretary
  - Message may be encrypted
- Because the envelope is marked as confidential, Office Assistant in office B passes it over to the confidential secretary

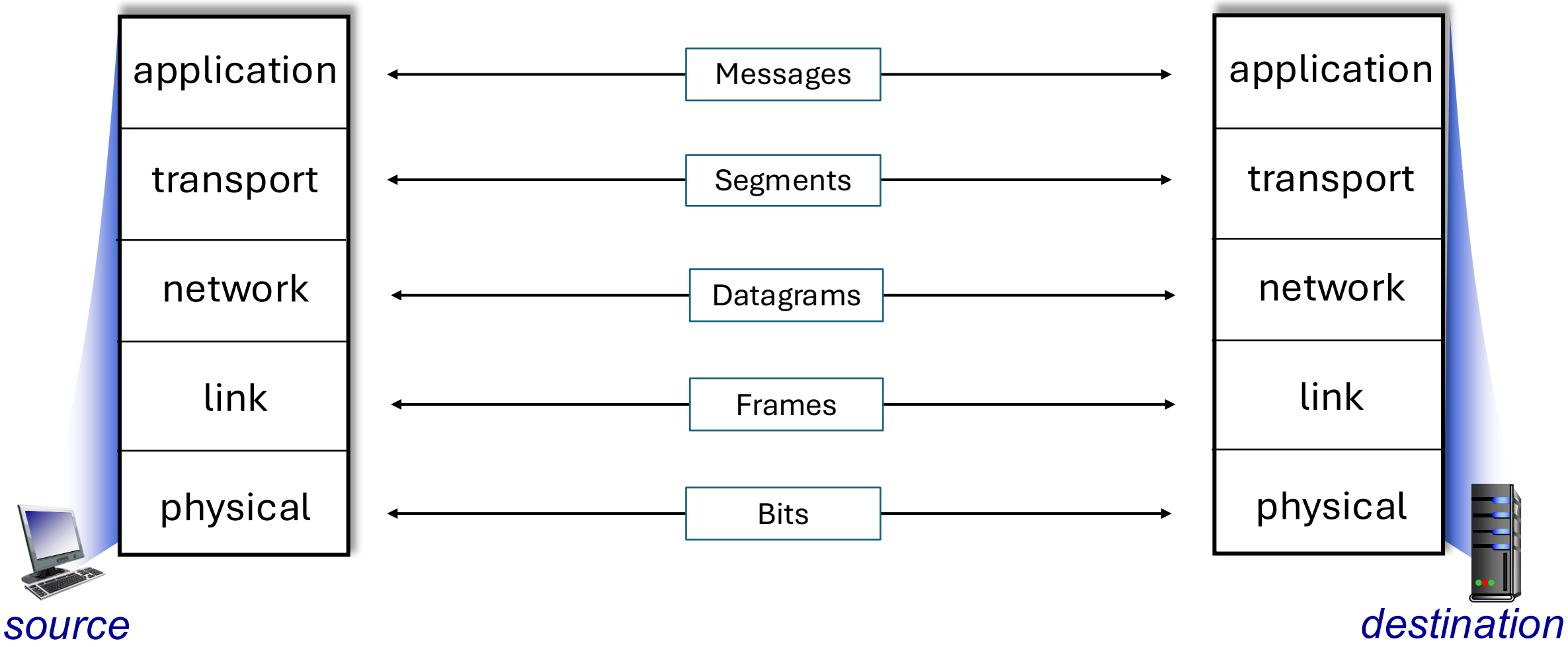


Layered Internet (TCP/IP) protocol stack

- *application:* supporting network applications
  - HTTP, IMAP, SMTP, DNS
- *transport:* process-process data transfer
  - TCP, UDP
- *network:* routing of datagrams from source to destination
  - IP, routing protocols
- *link:* data transfer between neighboring network elements
  - Ethernet, Wireless LAN (WiFi)
- *physical:* transporting bits "on the wire"



PDUs and peers

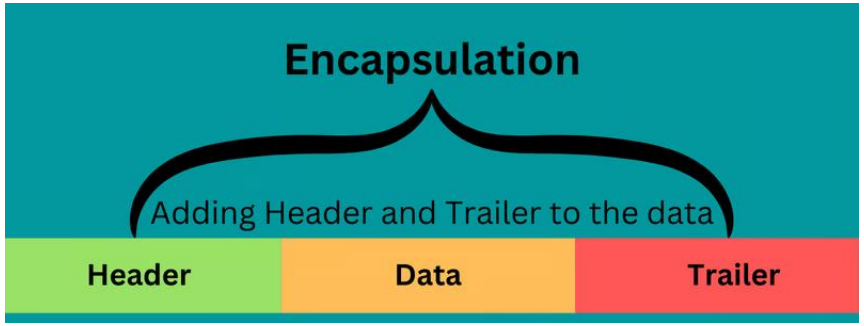


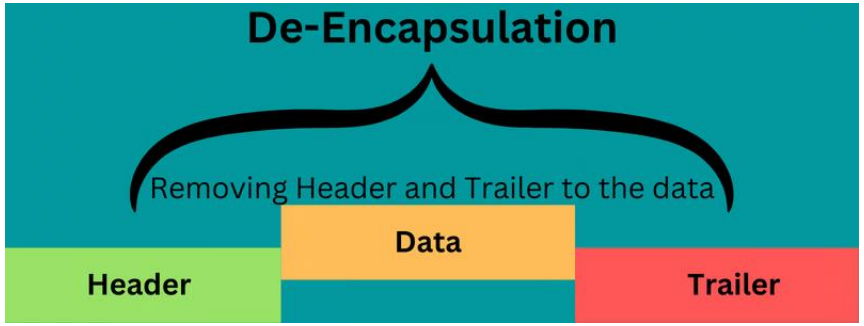
Encapsulation

- Encapsulation means putting in a cover or a container
  - Putting medicine in a capsule
  - Putting a letter in an envelope
- In networking, encapsulation is done by adding a header
  - In some cases a trailer as well
- De-encapsulation means removing the header (and trailer if present) to extract the original data





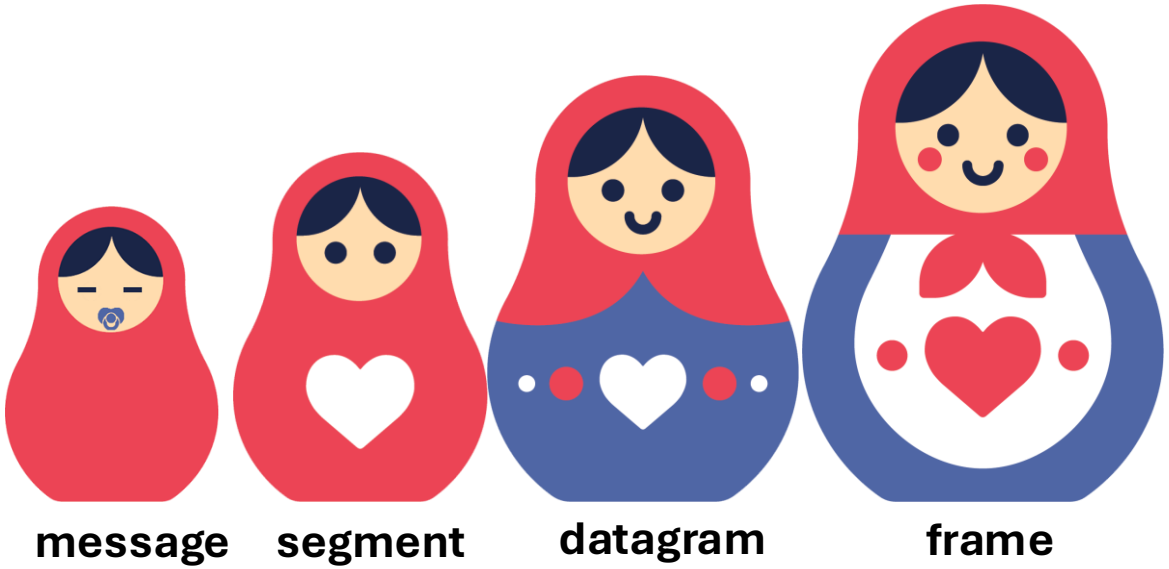




Encapsulation

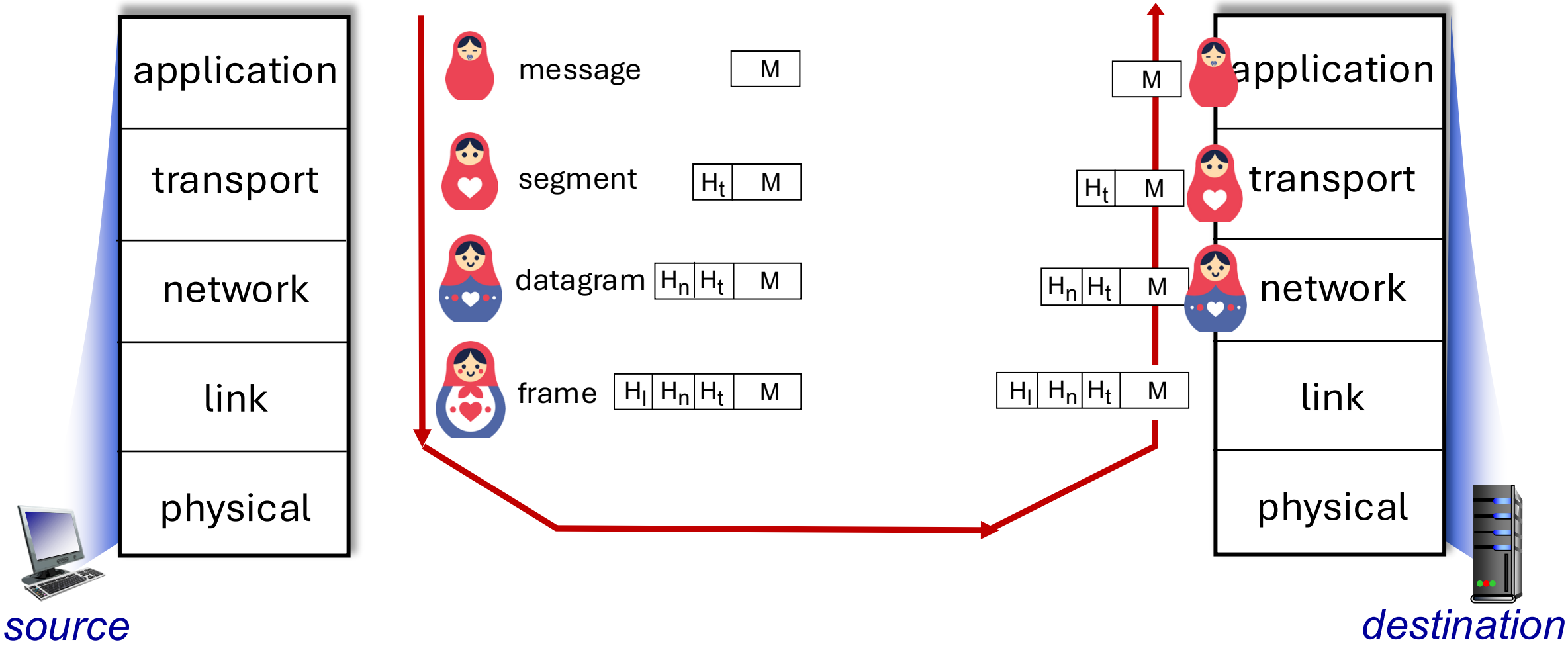
Matryoshka dolls (stacking dolls)



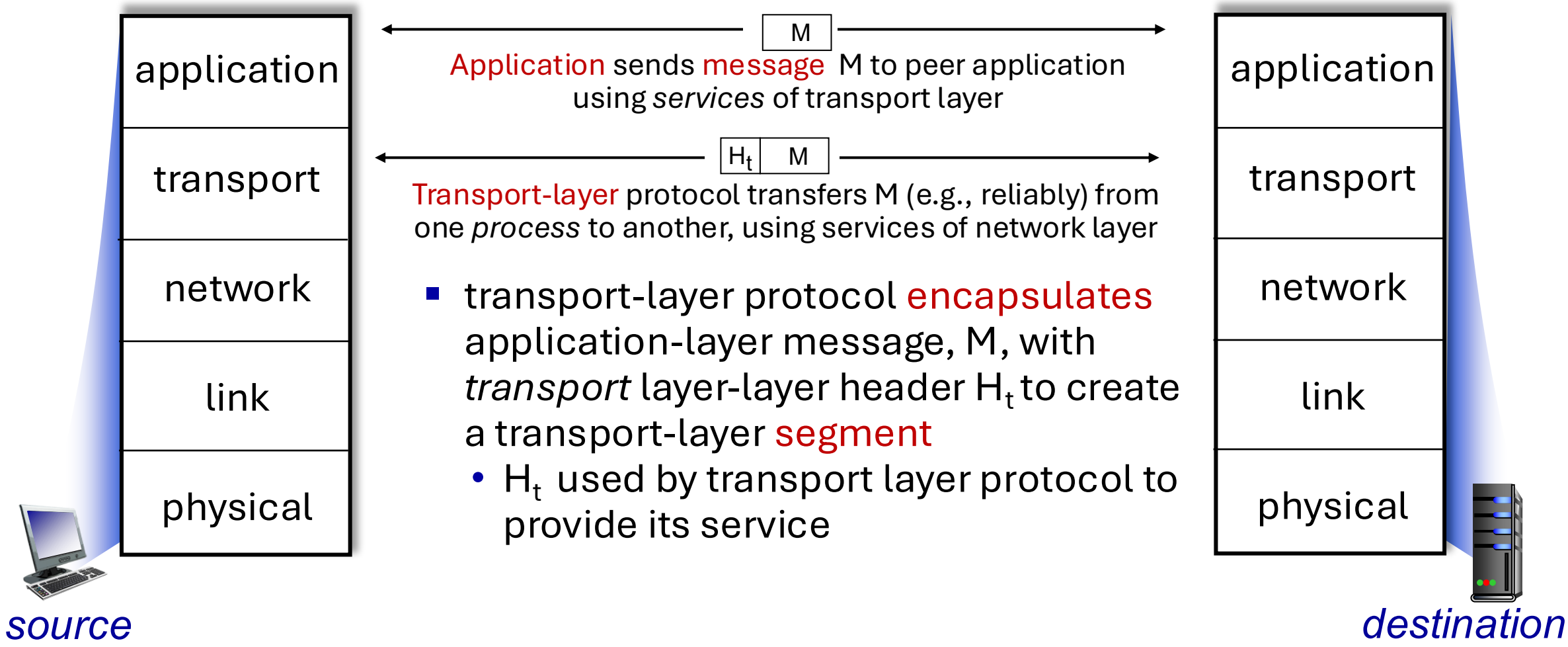


Credit: https://dribbble.com/shots/7182188-Babushka-Boi

Services, Layering and Encapsulation

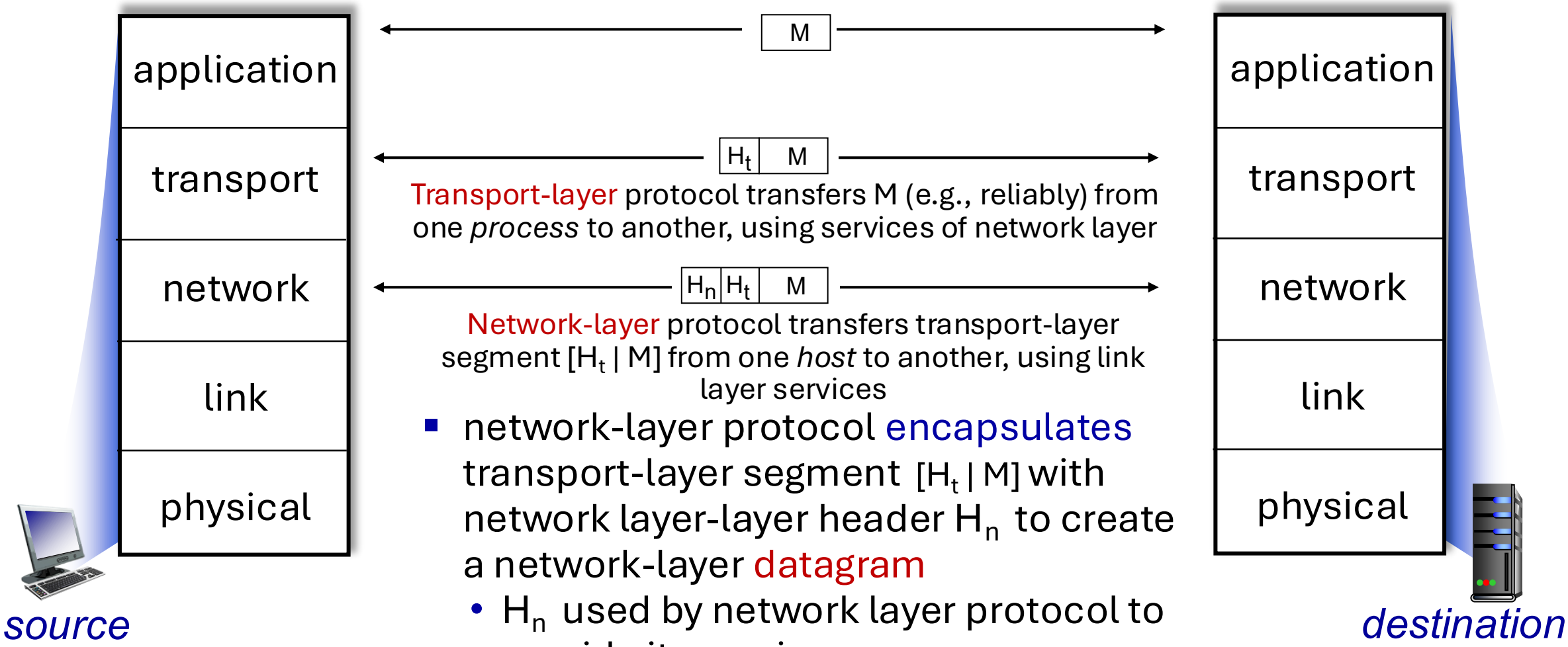


Services, Layering and Encapsulation



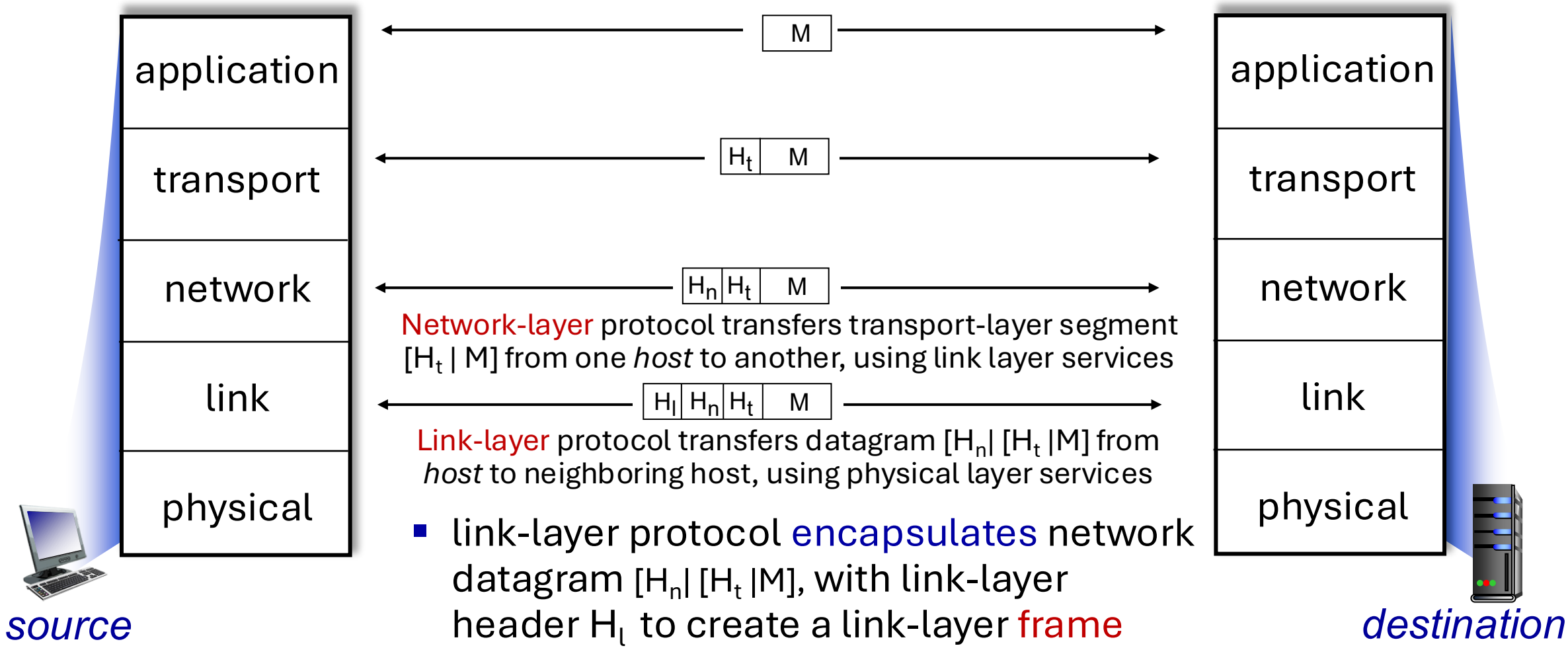
- transport-layer protocol encapsulates application-layer message, M, with *transport* layer-layer header H_t to create a transport-layer segment
  - H_t used by transport layer protocol to provide its service

Services, Layering and Encapsulation



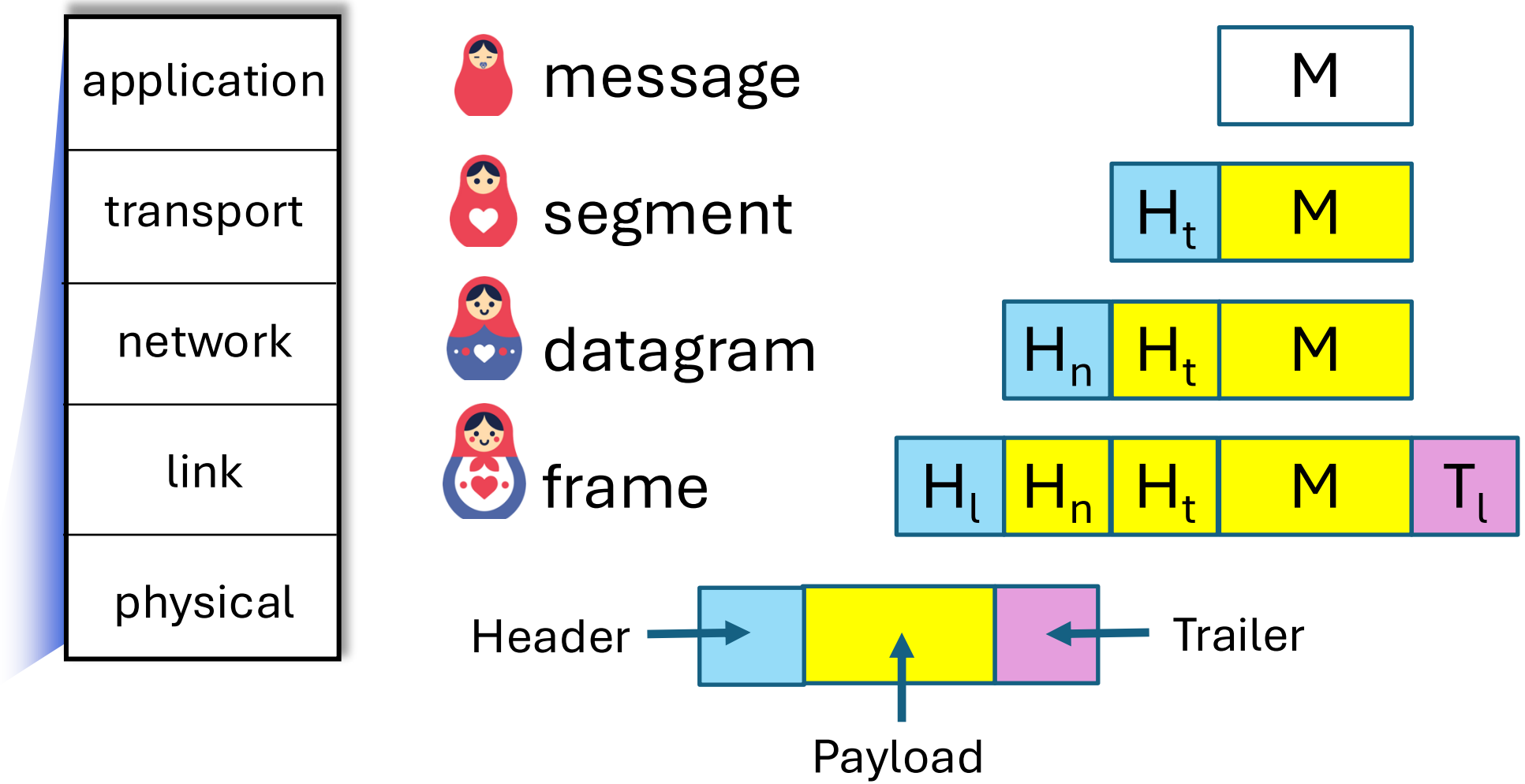
- network-layer protocol encapsulates transport-layer segment [H_t | M] with network layer-layer header H_n to create a network-layer datagram
  - H_n used by network layer protocol to provide its service

Services, Layering and Encapsulation



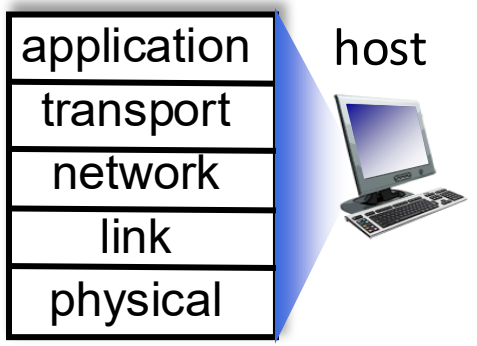
- link-layer protocol encapsulates network datagram [H_n| [H_t |M], with link-layer header H_l to create a link-layer frame

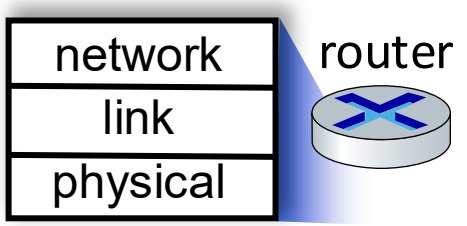
Packets: Headers, Trailers and Payloads

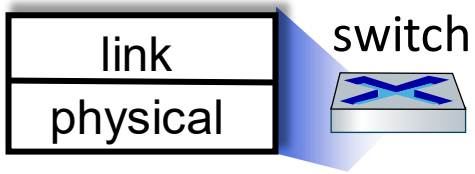


Layers present in devices

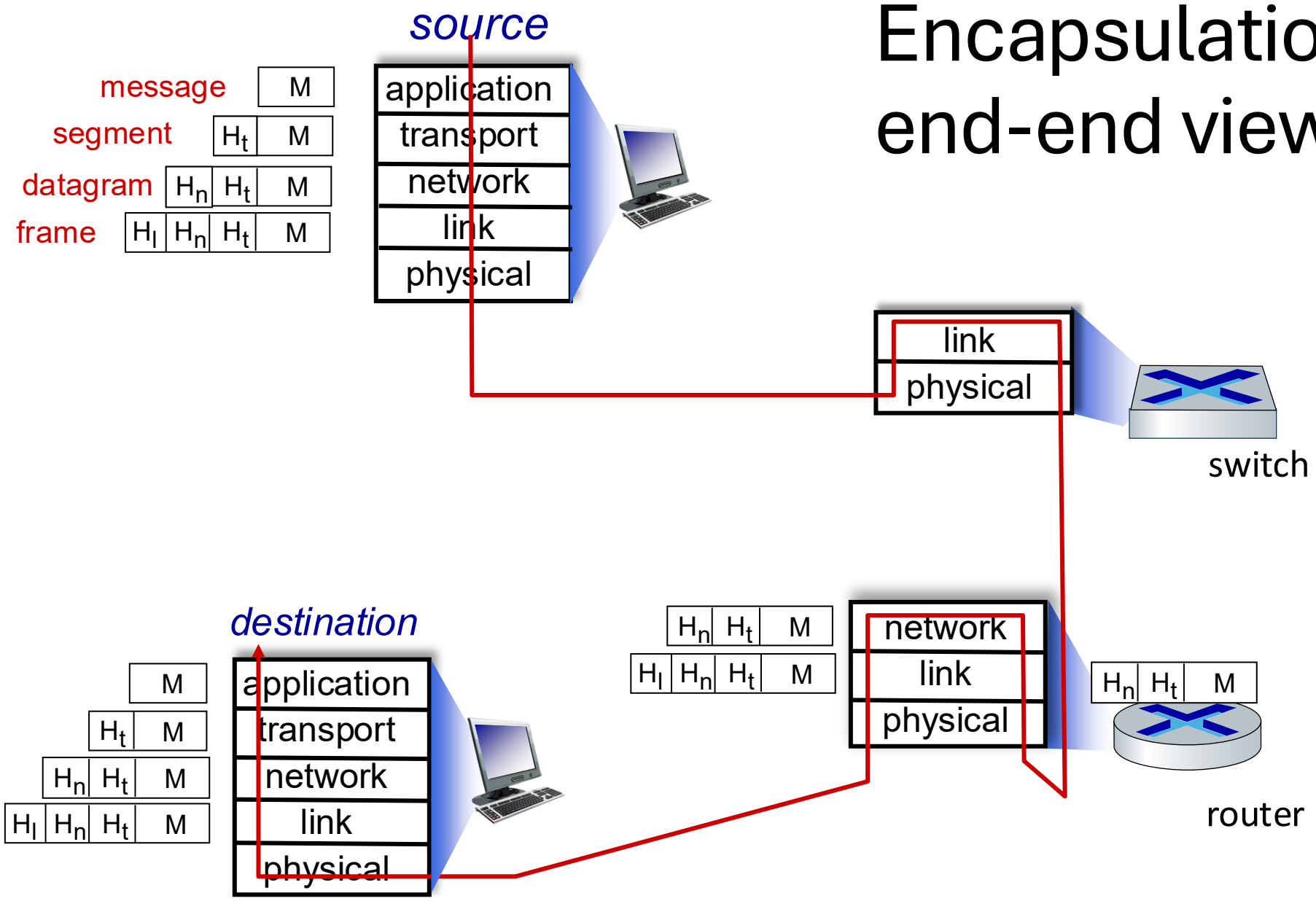
- Hosts (end-stations) have all 5 layers
- Routers have only the three lowest layers
  - Physical, Link and Network
  - They are called layer 3 devices or network layer devices for this reason
- Switches have only the two lowest layers
  - Physical and Link
  - They are called layer 2 devices or link layer devices for this reason
- Physical layer devices (like media converters) only have the physical layer







Encapsulation: an end-end view



TCP/IP layers: Application layer

- The application layer is where network applications and their application-layer protocols reside
  - Hyper Text Transfer Protocol: for web document request and transfer
  - Simple Mail Transfer Protocol: for the transfer of e-mail messages
  - Domain Name Service: for translation of a human-friendly name like courseweb.sliit.lk to a numerical network address
- The application in one host exchanges packets of information with the peer application in another host
  - following the rules of the specific application layer protocol
- We refer to this packet (PDU) of information at the application layer as a message

TCP/IP layers: Transport layer

- The Internet's transport layer is responsible for transporting application-layer messages between application endpoints
- In the transport layer of TCP/IP stack two main protocols are used
  - TCP – Transport Control Protocol
  - UDP – User Datagram Protocol
- TCP provides a connection-oriented, reliable service to applications
  - A connection needs to be established before any application data is sent
  - Reliability (guaranteed delivery) is provided through various methods
- UDP provides a connectionless service to applications
  - No need to establish a connection before sending data
  - Provides no reliability
- A transport-layer packet (PDU) as a segment

TCP/IP layers: Network layer (IP layer)

- The network layer is responsible for moving network-layer packets known as datagrams from one host to another
- The transport-layer in the source host passes a transport-layer segment and a destination address to the network layer
  - just like destination address of a letter that you post using the postal service
- The network layer has to deliver the segment to the transport layer in the destination host
- The main protocol used in the network layer is the Internet Protocol (IP)
  - Two versions: IPv4 and IPv6
- This layer is also called IP layer, as IP is the main protocol used
- IP layer also has a number of other protocols

TCP/IP layers: Link layer

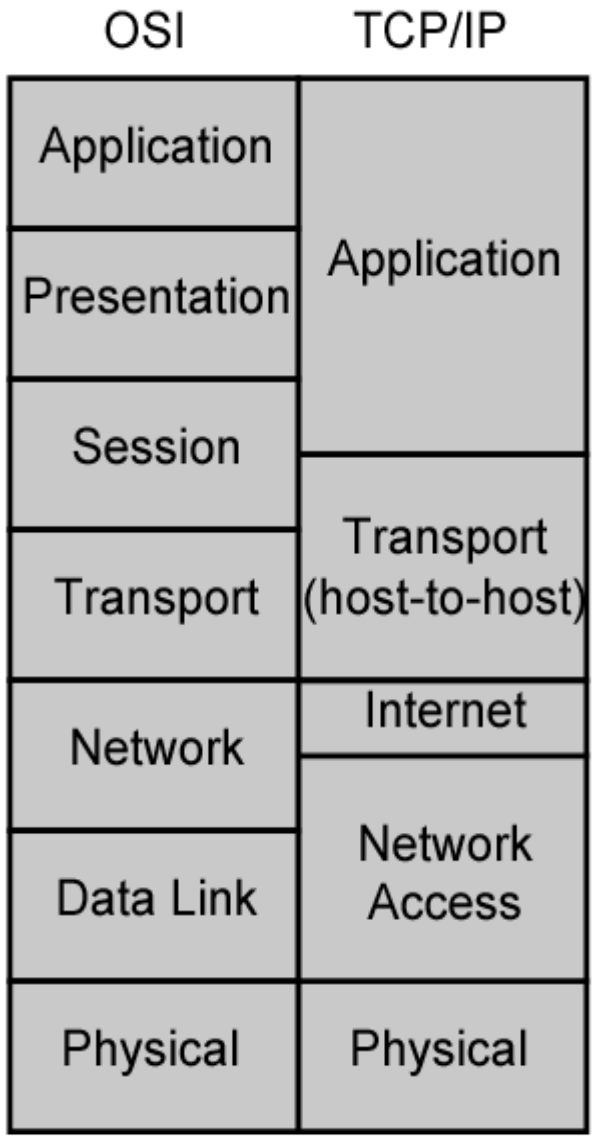
- The IP layer routes a datagram through a series of routers between the source and destination
- To move a packet from one node (host or router) to the next node in the route, the network layer relies on the services of the link layer
- At each node, the network layer passes the datagram down to the link layer, which delivers the datagram to the next node along the route
- At this next node, the link layer passes the datagram up to the network layer
- Examples of link-layer protocols include
  - Ethernet: for wired local area networks (LANs)
  - Wi-Fi: wireless LANs
- We refer to the link-layer packets (PDUs) as frames

TCP/IP layers: Physical layer

- The job of the physical layer is to move the individual bits (1s and 0s) in the link layer frame from one node to the next
- The protocols in this layer depend on the actual transmission medium of the link
- For example, Ethernet has many physical-layer protocols:
  - one for twisted-pair copper wire
  - another for optical fiber cable
  - …
- In each case, a bit is moved across the link in a different way

ISO / OSI reference model

- Open Systems Interconnect (OSI) architecture developed by ISO (International Standards Organization)
- It is more of a “reference” architecture as opposed to the TCP/IP stack which is based on actual protocols in use
- Has a total of 7 layers, compared to the 5 layers of TCP/IP stack
  - Additional layers are Session and Presentation
  - In the TCP/IP stack the functions of these layers are performed by Transport and Application layers



Review questions

- What are the principal responsibilities of each of the layers in the TCP/IP stack?
- What are the names used to refer to PDUs in
  - Application layer
  - Transport layer
  - Network layer
  - Link layer
- Which layers of the TCP/IP stack can these devices process
  - A host
  - A switch
  - A router

References

- Chapter 1 - Introduction
  James F. Kurose and Keith W. Ross, Computer Networking – A Top-Down Approach, (8th Edition), Pearson, 2020
- Chapter 2 - Protocol Architecture, TCP/IP, and Internet-Based Applications
  William Stallings, Data and Computer Communications (10th Edition), Pearson, 2014

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